VEHICLE DYNAMICS AND CONTROL- ADAMS LAB

This report details the suspension and full vehicle analyses performed using MSC Adams Car. A parametric study is carried out to evaluate the effects of various suspension and vehicle parameters on overall vehicle dynamics and handling performance.

ADAMS CAR - SUSPENSION

The analysis is carried out on an existing suspension assembly of the Ferrari TestaRossa in the shared folder of ADAMS . The Front Double Wishbone Suspension and Steering assembly are considered.

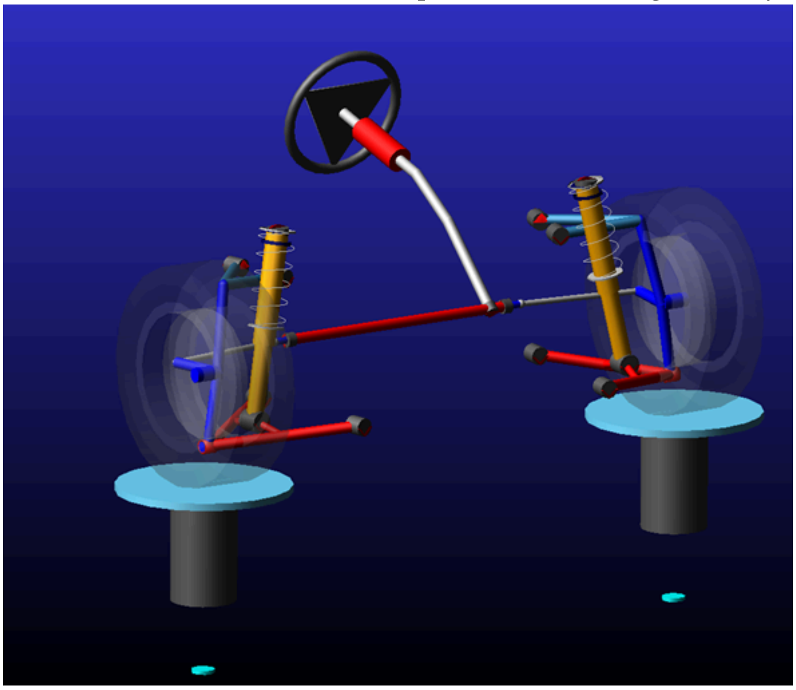
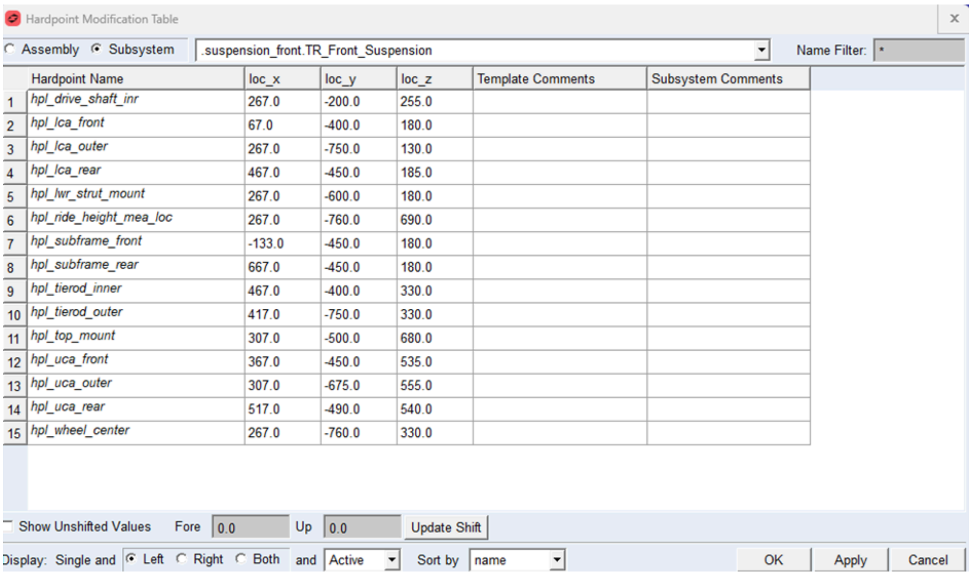


Fig 1 - Front Suspension Assembly in ADAMS Car

To carry out the suspension analysis, the suspension geometry is varied by altering the hardpoints and their effect on various kinematic indicators is discussed.



Hardpoint Modification Table

Assembly | Subsystem | .suspension_front.TR_Front_Suspension | Name Filter: *

	Hardpoint Name	loc_x	loc_y	loc_z	Template Comments	Subsystem Comments
1	hpl_drive_shaft_inr	267.0	-200.0	255.0		
2	hpl_lca_front	67.0	-400.0	180.0		
3	hpl_lca_outer	267.0	-750.0	130.0		
4	hpl_lca_rear	467.0	-450.0	185.0		
5	hpl_lwr_strut_mount	267.0	-600.0	180.0		
6	hpl_ride_height_mea_loc	267.0	-760.0	690.0		
7	hpl_subframe_front	-133.0	-450.0	180.0		
8	hpl_subframe_rear	667.0	-450.0	180.0		
9	hpl_tierod_inner	467.0	-400.0	330.0		
10	hpl_tierod_outer	417.0	-750.0	330.0		
11	hpl_top_mount	307.0	-500.0	680.0		
12	hpl_uca_front	367.0	-450.0	535.0		
13	hpl_uca_outer	307.0	-675.0	555.0		
14	hpl_uca_rear	517.0	-490.0	540.0		
15	hpl_wheel_center	267.0	-760.0	330.0		

Show Unshifted Values | Fore 0.0 | Up 0.0 | Update Shift

Display: Single and | Left | Right | Both | and Active | Sort by name | OK | Apply | Cancel

Fig 2 - Hardpoint Modification Table

a. Parallel Wheel Travel Analysis-

In the first stage of the suspension analysis, a parallel wheel travel study is performed to evaluate the suspension kinematics. Key parameters, including camber and toe variations throughout the wheel travel range, are analysed and discussed to assess their influence on vehicle handling and tire performance.

- Camber Gain vs Wheel Travel

  The upper control arm inboard mounting points (hpl_uca_front and hpl_uca_rear) are modified to sweep them in a range of +-20mm and the effect on camber gain is studied.

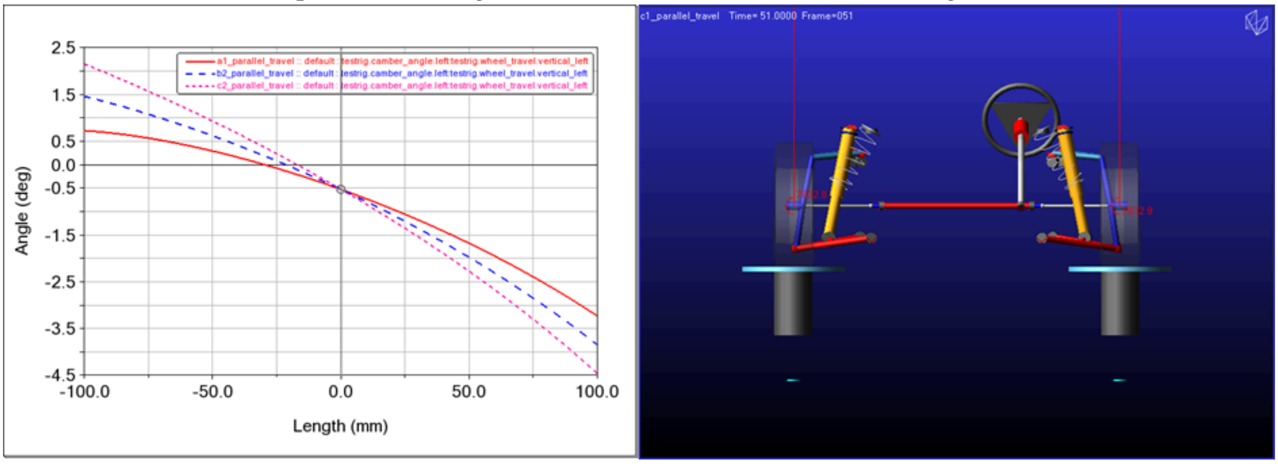


Fig.3 - ADAMS Car Post-Processing

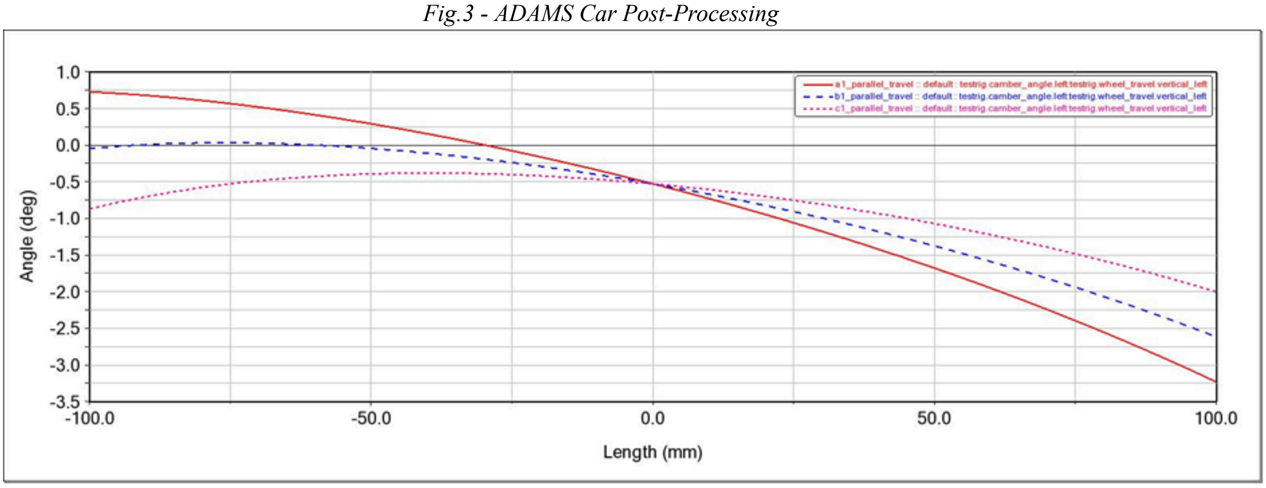


Fig.4 - Camber Gain Variation on increasing UCA mounting points

In Fig.4, the red line denotes the baseline camber gain. The plot for the baseline analysis shows a camber change from +0.8° to -3.3° for a 100mm wheel travel in bump and rebound.

On increasing the vertical position of the mounting points by 10mm and 20mm, the camber gain reduces and on increasing it even further it starts to show very little variation. The camber curves become flatter compared to the baseline configuration, indicating a smaller change in wheel camber for a given suspension displacement. This reduces the generation of negative camber during bump, which may improve kinematic stability but can compromise cornering grip by providing less camber compensation during vehicle roll.

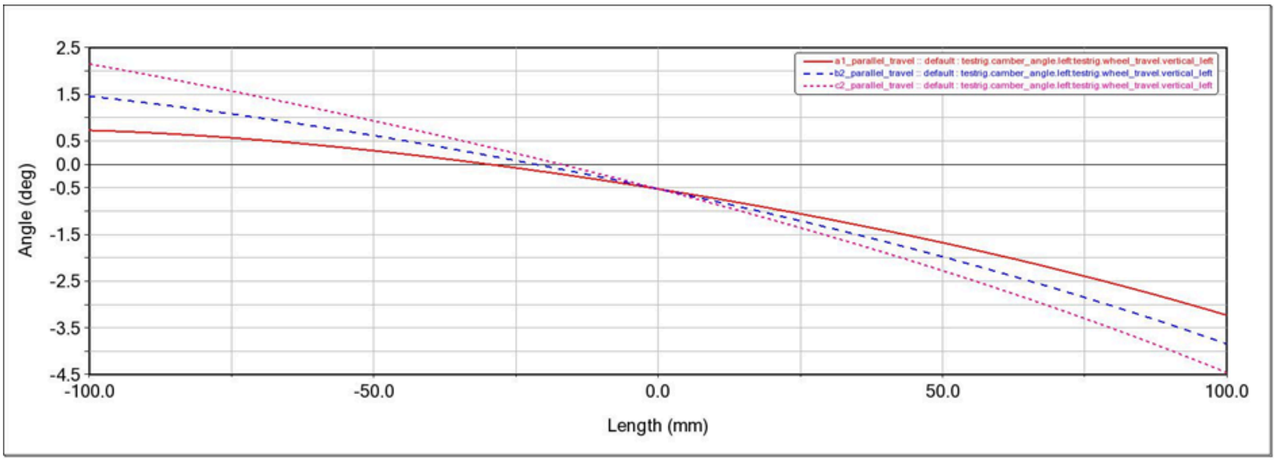


Fig.5 - Camber Gain Variation on decreasing UCA mounting points

Fig.5 shows the camber gain curve on decreasing the vertical height of the mounting points by 10mm and 20mm. The behaviour is exactly opposite to what was seen before.

Compared with the baseline configuration, the modified geometries produce larger positive camber in rebound and larger negative camber in bump, resulting in steeper camber curves. This behaviour enhances camber compensation during vehicle roll and can improve cornering performance; however, it also increases the sensitivity of wheel camber to suspension displacement and may lead to higher tire wear.

- Toe vs Wheel Travel
  The effect of varying the lateral positioning of the outer tie rod mounting by +-30mm is discussed.

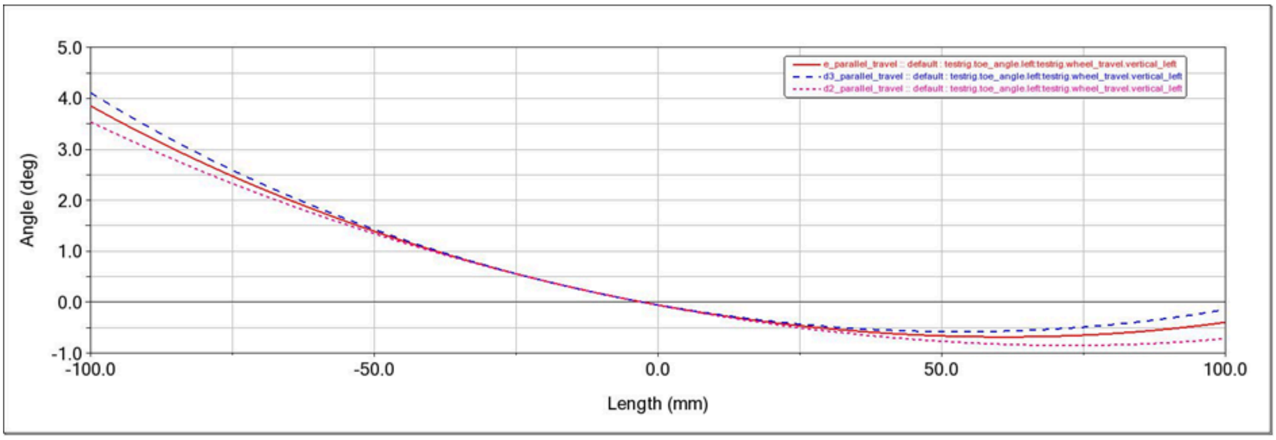


Fig.6 - Toe Variation on varying Tie Rod outer mounting points

Varying the outer tie rod mounting position by ±30 mm significantly influences the bump steer characteristics of the suspension. Lowering the mounting point by 30 mm reduces the magnitude of toe variation throughout wheel travel, producing a more stable toe response. In contrast, increasing the mounting position by 30 mm results in greater toe changes at both bump and rebound extremes, indicating increased bump steer sensitivity. The baseline configuration exhibits an intermediate response between the two modified geometries.

b. Dynamic Analysis

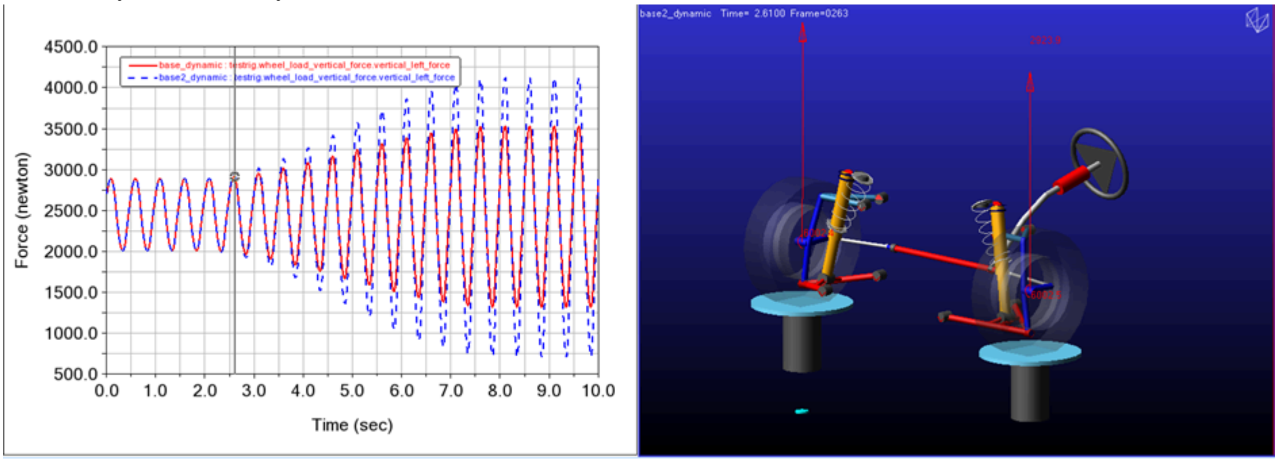


Fig.7 - Dynamic Analysis of Suspension in Adams Car Postprocessor

- Wheel Vertical Contact Force vs Time
  The input road amplitude is made to vary periodically from 10mm to 30mm for the base analysis depicted by the red curve in Fig.8.
  Then the peak amplitude is varied to go till 50mm and we can see the vertical contact forces also increase respectively.

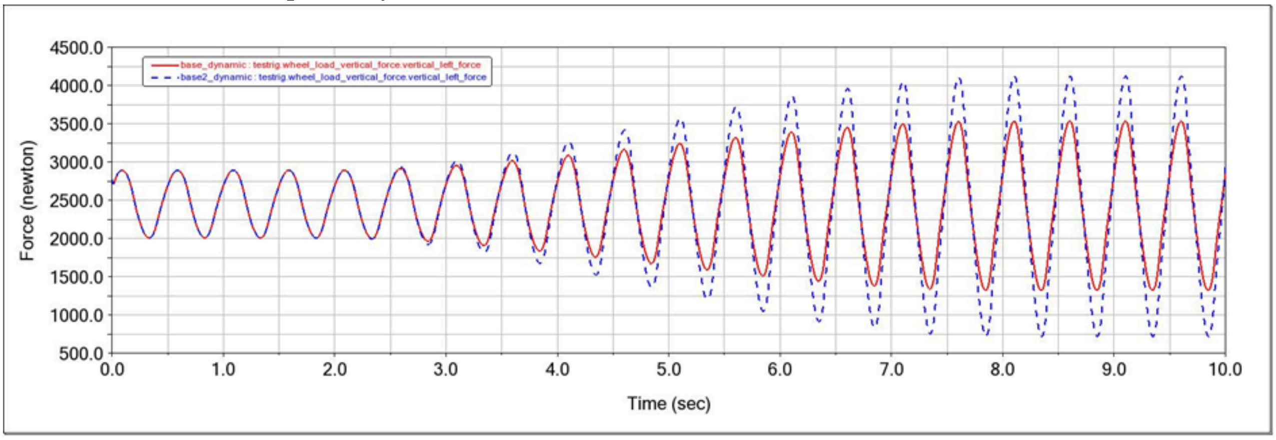


Fig.8 - Wheel Vertical Forces vs Time

- Bushing Force vs Time
  Similarly, the same input change is used to plot the bushing force variation with time as follows.

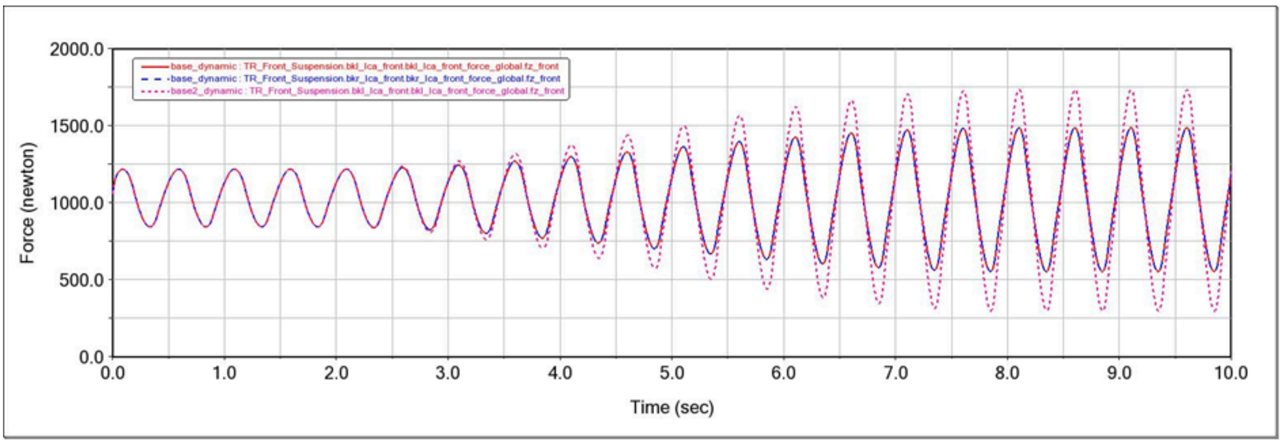
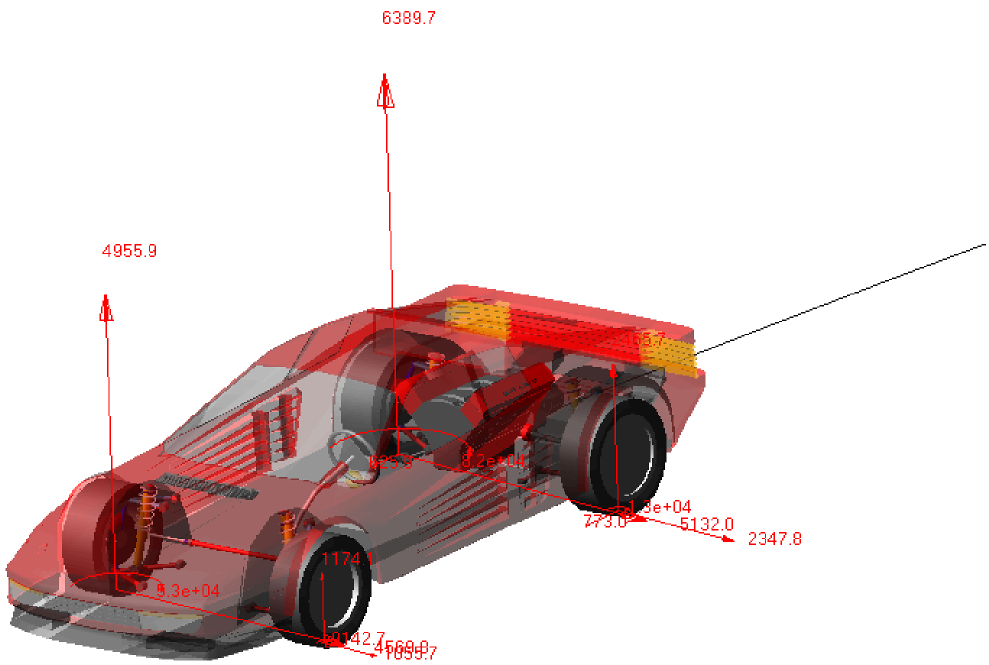
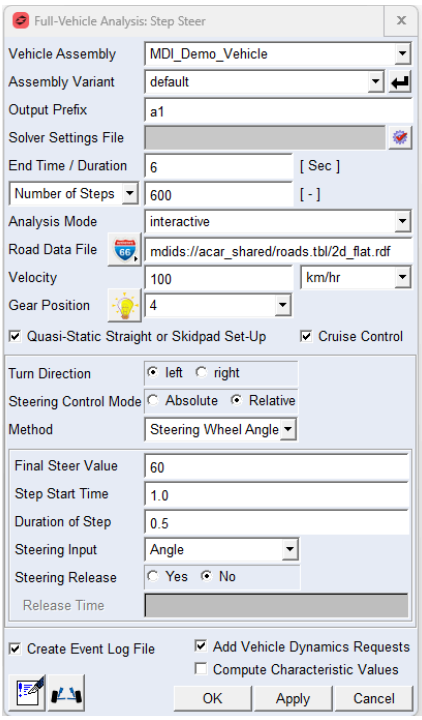


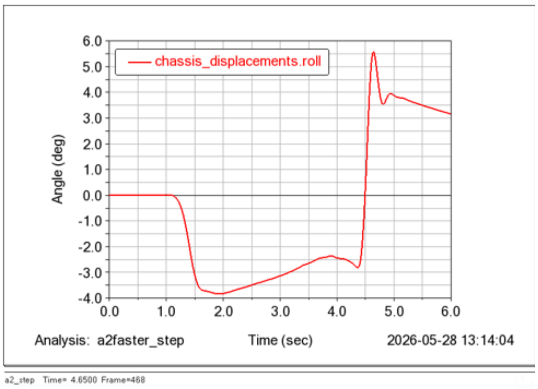
Fig.9 - Bushing Forces vs Time

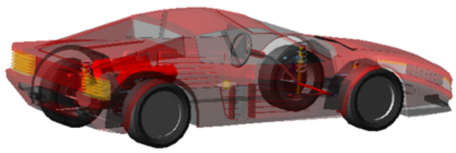
ADAMS CAR - FULL VEHICLE

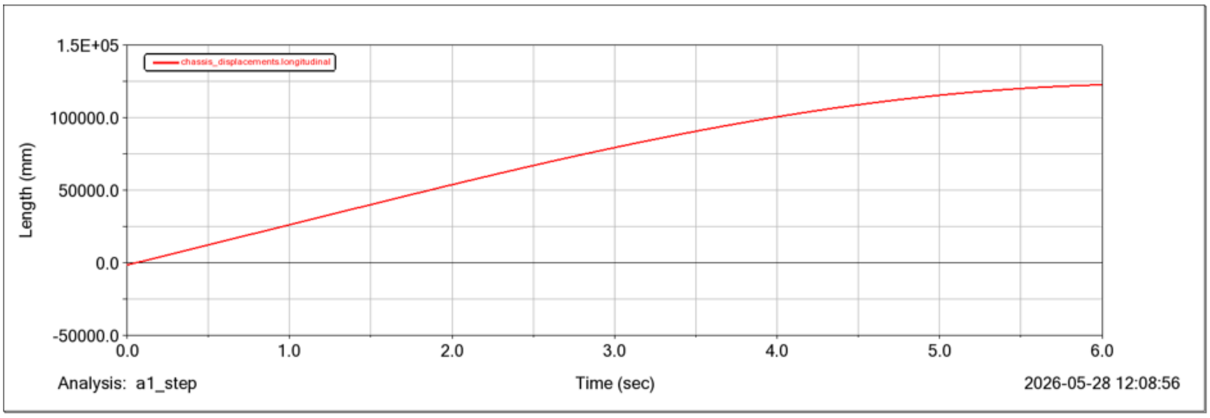


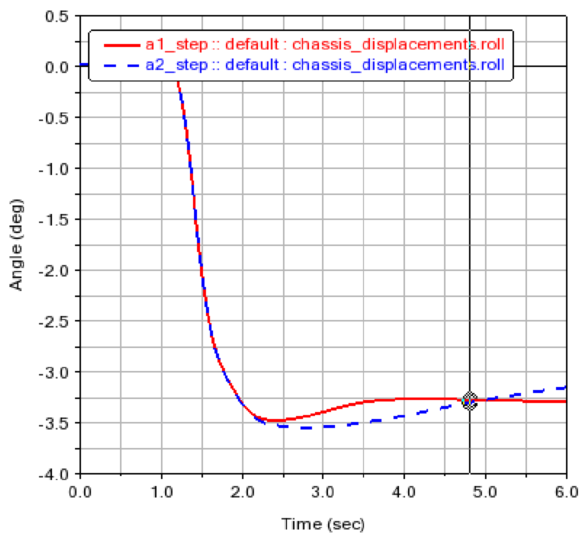


A. Fullbody Simulation

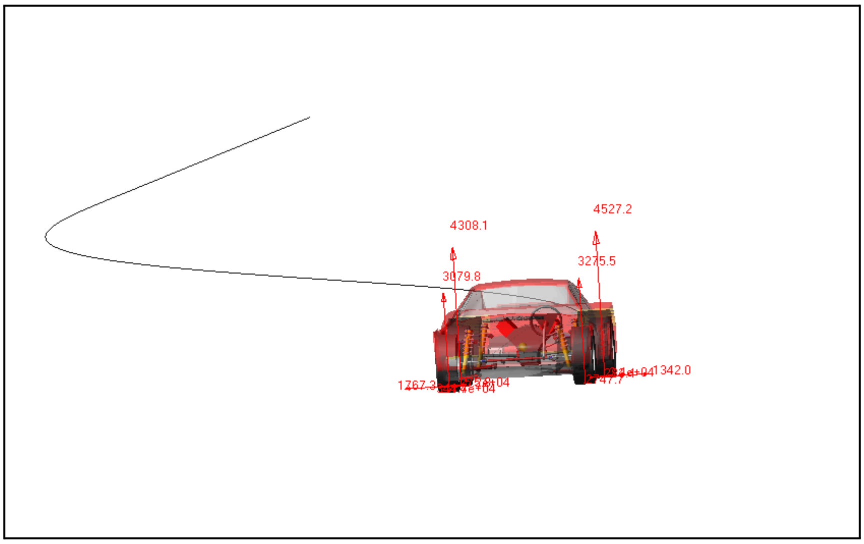


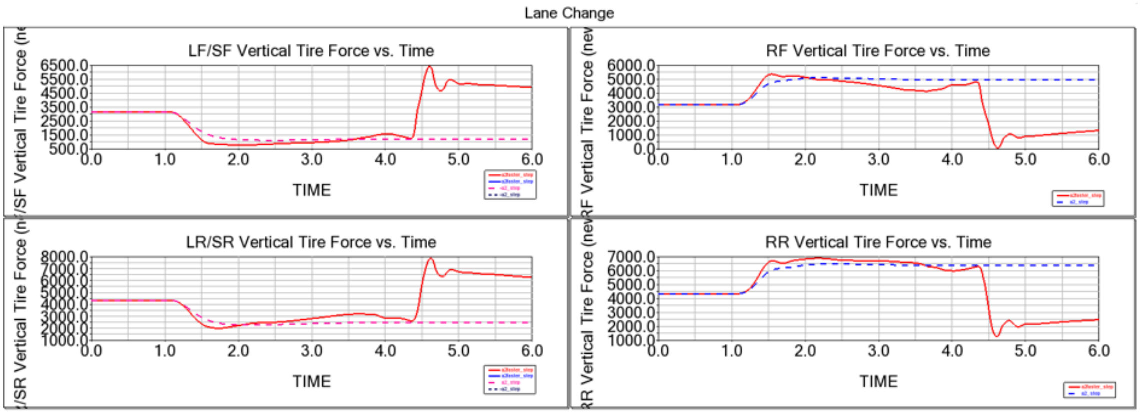


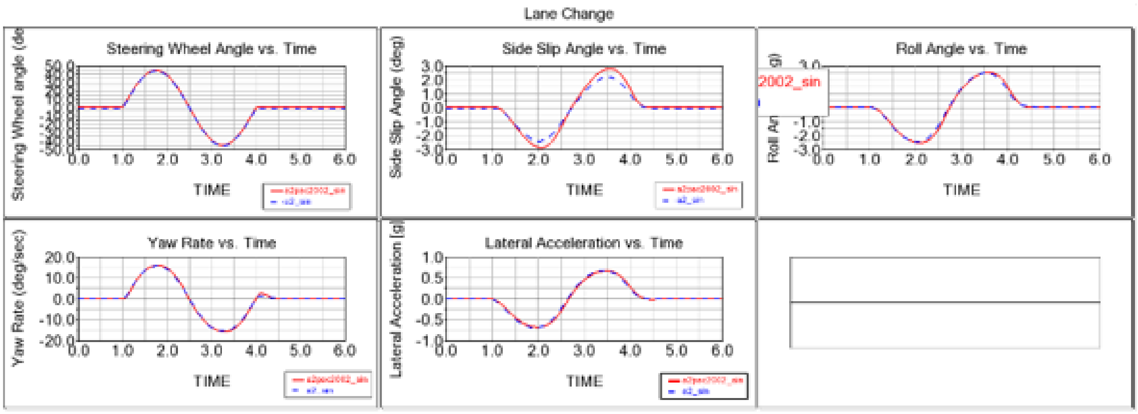


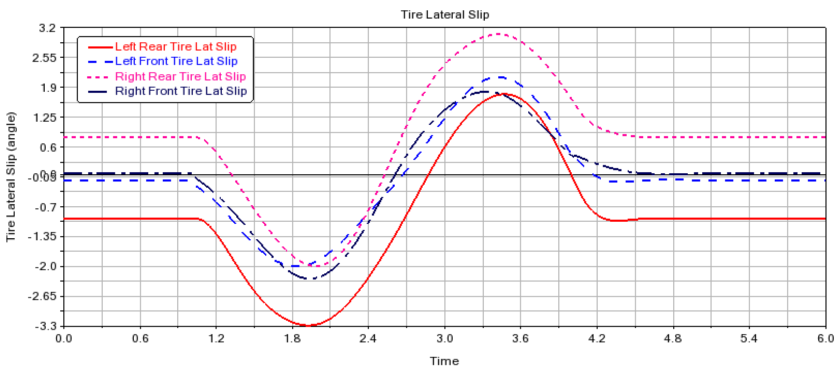


B. Lane Change

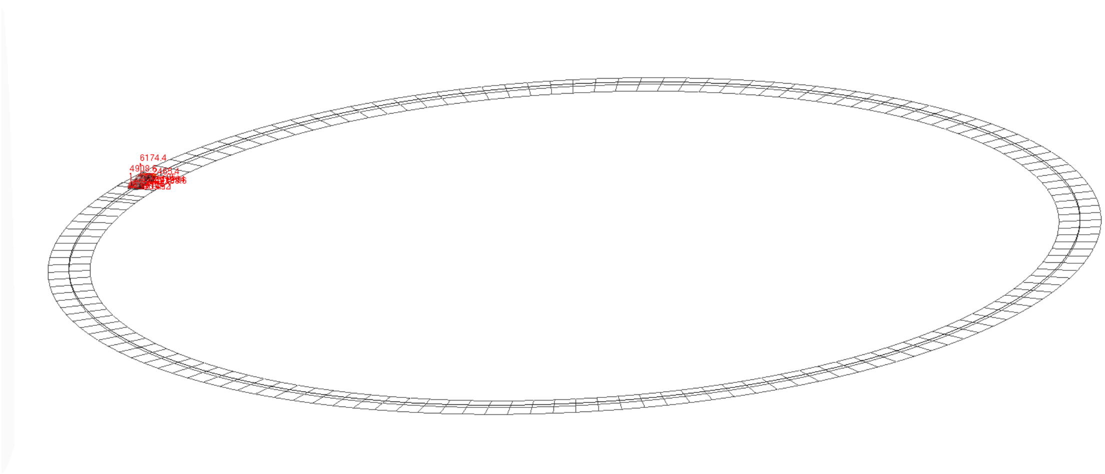


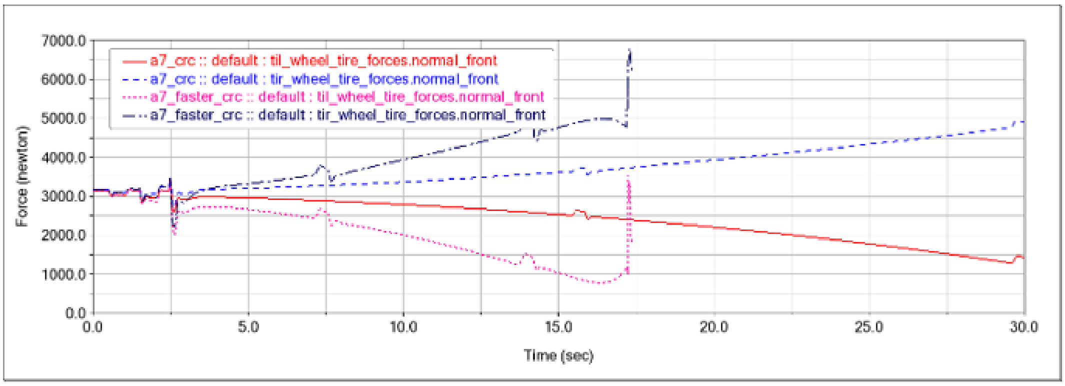


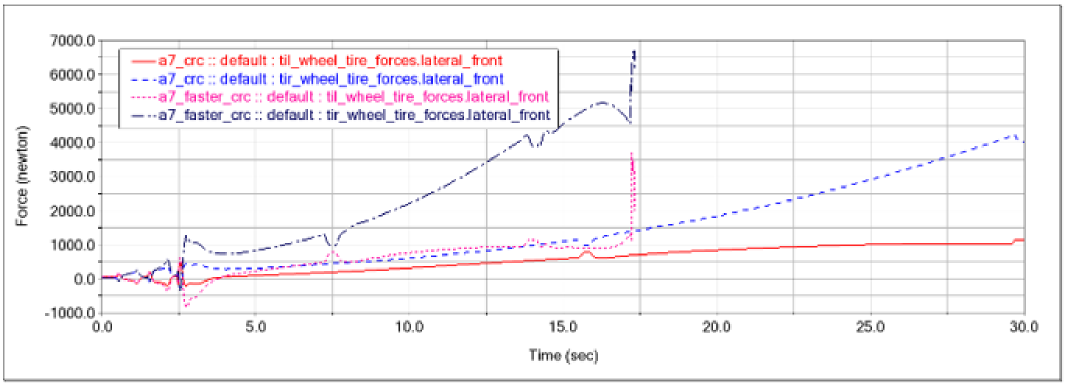


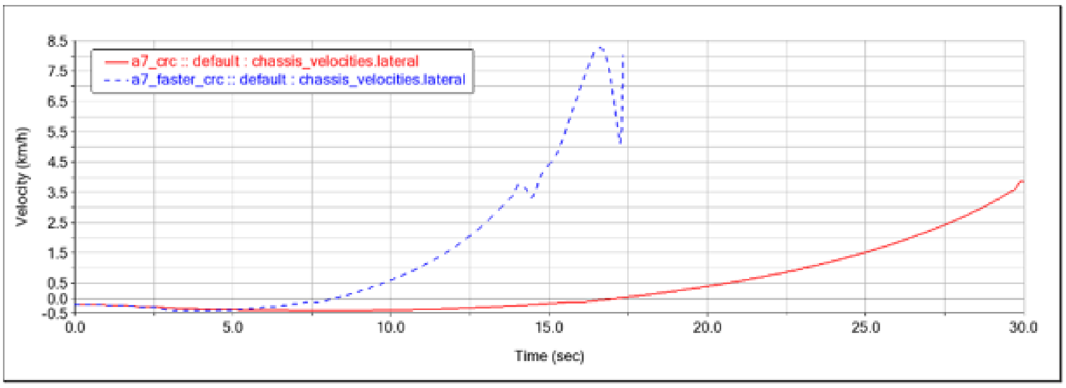


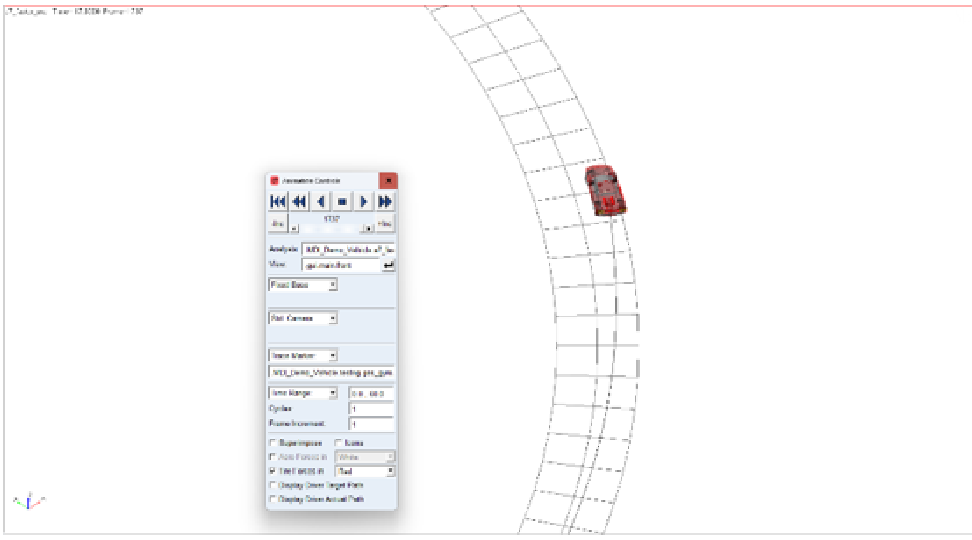
C. Constant Radius Cornering

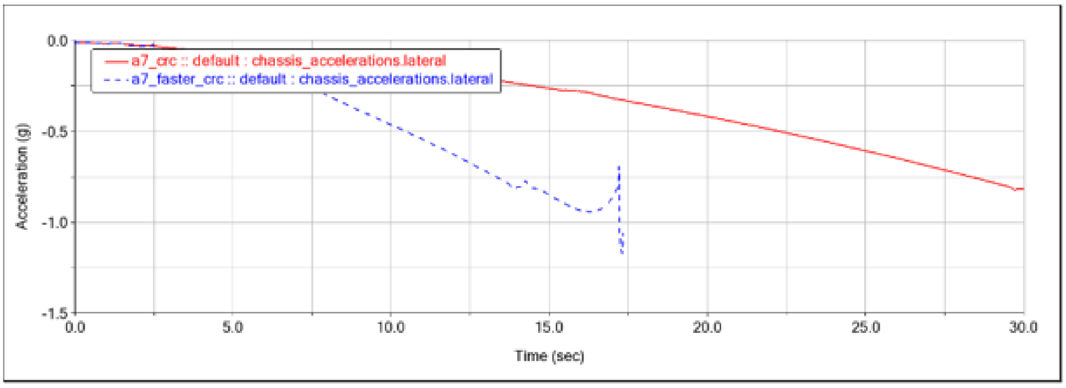


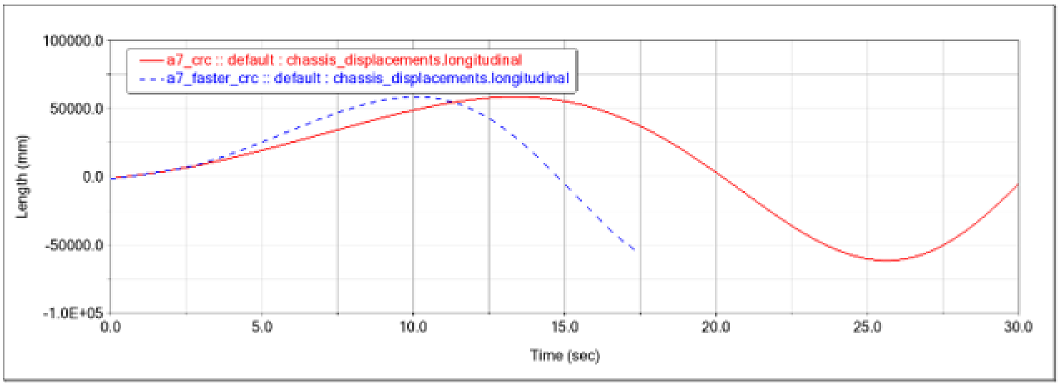






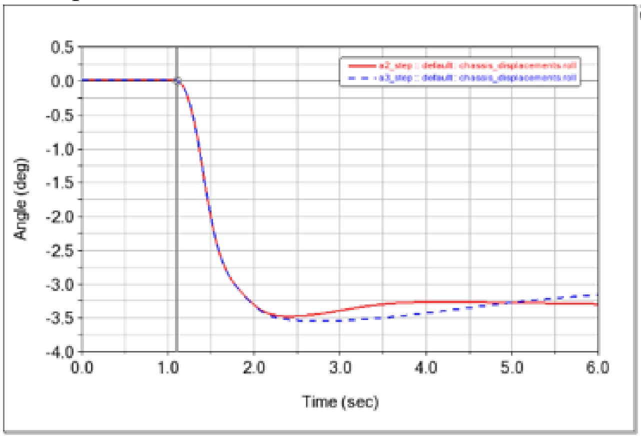


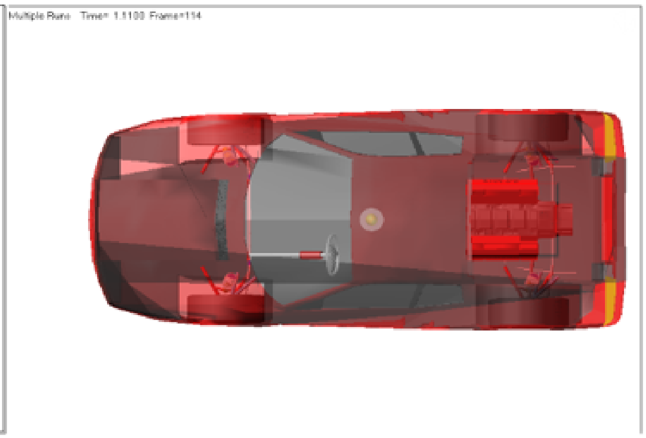


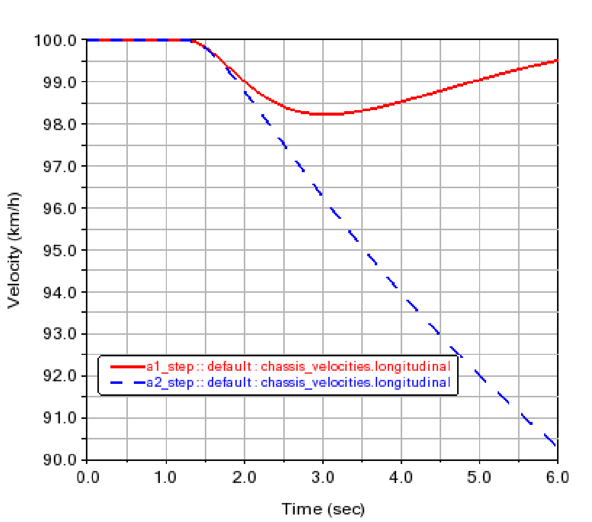


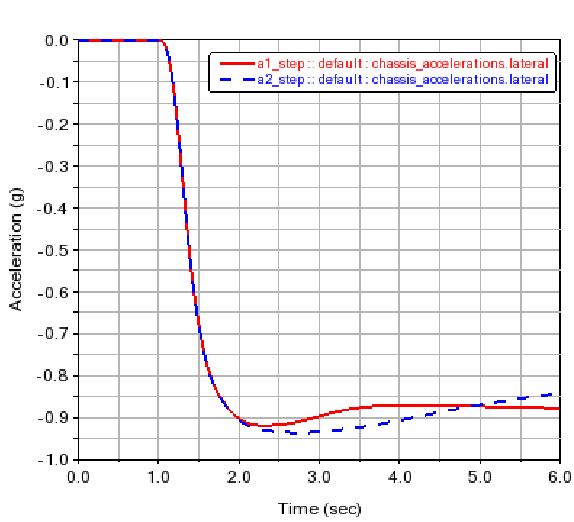
The constant radius cornering condition is simulated and two cases are compared. For the case concerning the higher speed, the vehicle loses grip at the front wheels tends and towards heavy understeer and leaves the track as highlighted by the graphs.

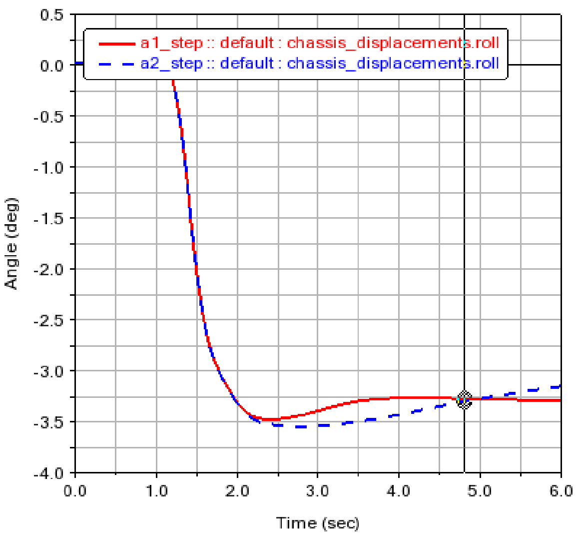
D. Step Steer





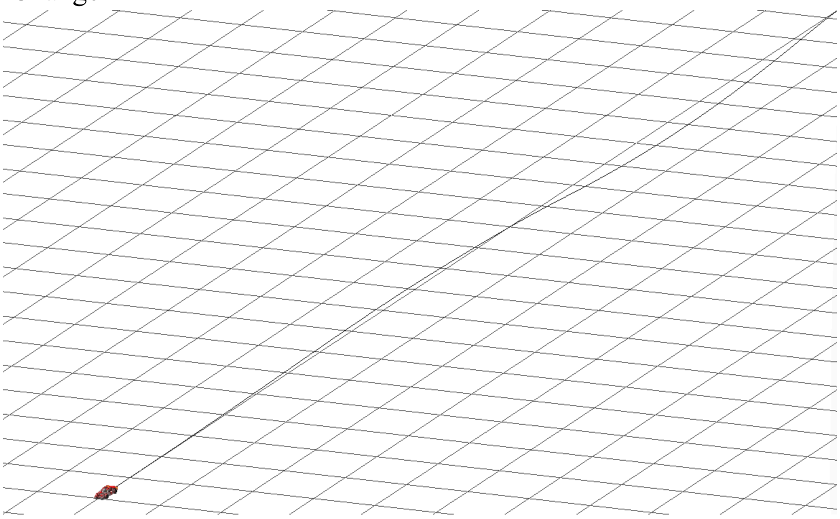


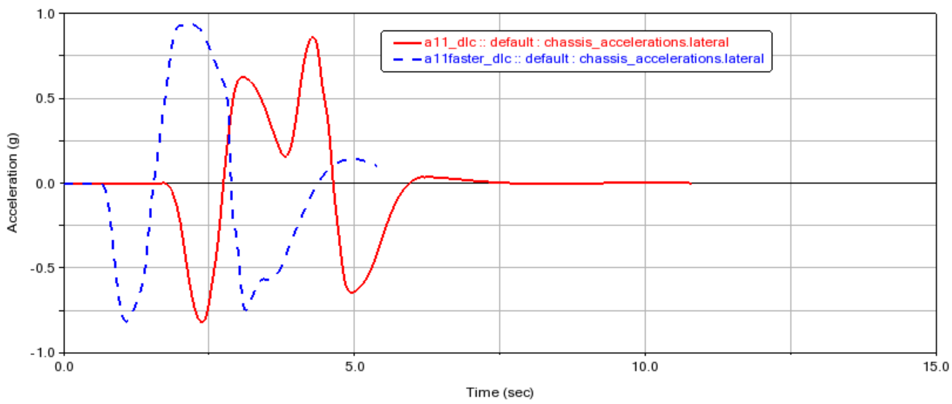


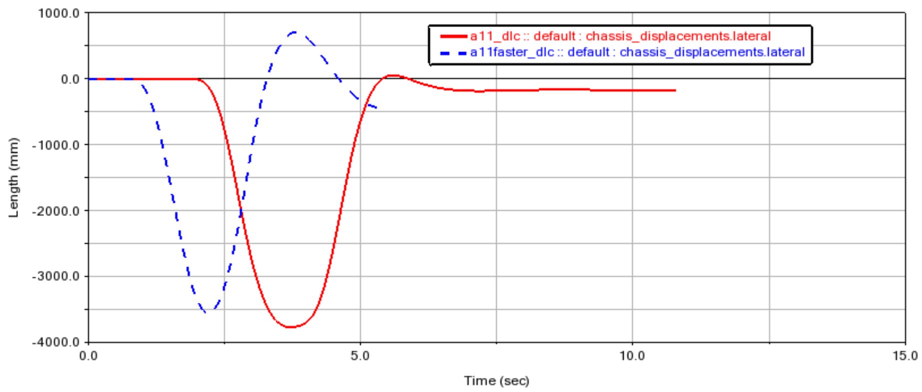


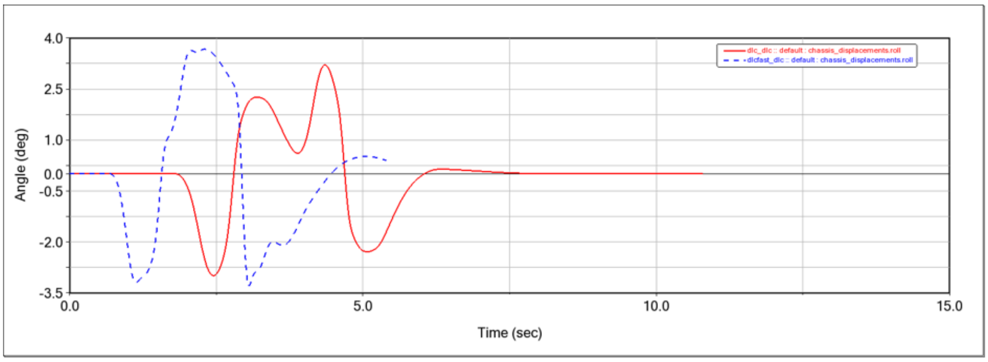
The step steer condition is simulated. Comparison is made on two cases having different velocities.

E. Double Lane Change

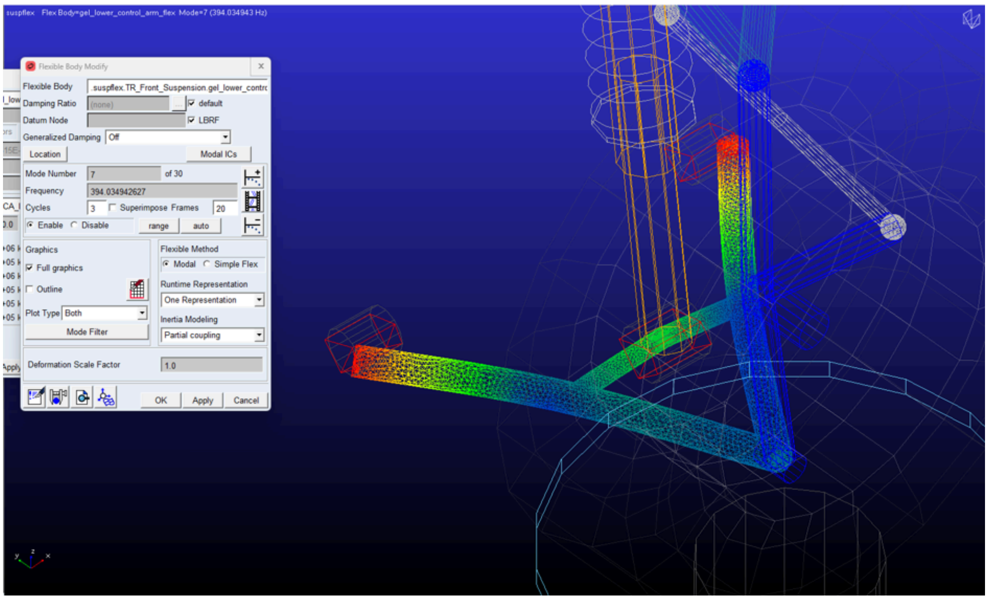


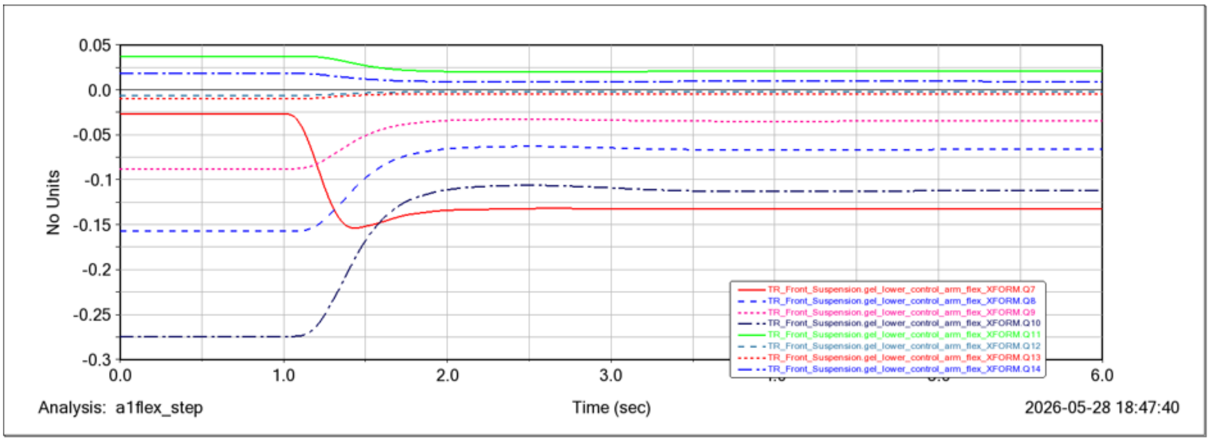






F. Flexbody





In the simulations performed in ADAMS for the fullbody vehicle, we have simulated for various scenarios and plotted various responses such as chassis roll, lateral acceleration, chassis displacement etc. Various comparisons are shown by changing vehicle parameters like suspension hardpoints and vehicle speed.

Furthermore, the flexbody simulation highlights the various modal excitations that are experienced by the lower control arm when it is not considered as a rigid body. The modal contributions from mode 7 to 14 are highlighted.